

Differential Equations, Built From Scratch

A ground-up companion to the Lecture 1 slides — every idea explained in plain language, with a few original worked examples along the way.

0 The One Idea Everything Builds On

TL;DR

A derivative is just a rate of change — that one idea is the foundation for everything else here.

You already know what a derivative is: it's a **rate of change**. If y depends on x , then $\frac{dy}{dx}$ just answers "if x nudges up a tiny bit, how much does y nudge?" That's the whole idea — a derivative is a speed of change, whatever the two quantities happen to be.

1 The Big Mental Shift: Solving for a Function, Not a Number

TL;DR

In algebra you solve for a number. Here, you solve for an entire function.

In every algebra course, "solve for x " meant: find a *number*. $2x + 1 = 7 \Rightarrow x = 3$. One number, done.

A differential equation asks something different. It gives a clue about how something *changes*, and the task is to reconstruct the whole story: the function itself.

LET'S DERIVE ONE FROM SCRATCH

Imagine a fast-growing plant, described this way: "its height grows at a rate equal to twice its current height, in cm per week." In symbols:

$$\frac{dh}{dt} = 2h$$

This isn't a number for the height — it's a *rule* the height obeys at every instant. The task is to find the function $h(t)$, the entire growth curve, that obeys this rule.

It turns out $h(t) = h_0 e^{2t}$ (where h_0 is the starting height) does the job. Check it:

$$\frac{d}{dt} (h_0 e^{2t}) = 2h_0 e^{2t} = 2h(t) \checkmark$$

You're given a relationship between a function and its own rate(s) of change — and the goal is to find the function.

2

Where These Equations Actually Come From

TL;DR

Differential equations come from describing behavior in plain language, then translating that description into symbols.

Coffee cooling down

Think about a hot cup of coffee sitting in a room. From experience:

- When the coffee is much hotter than the room, it loses heat fast.
- As it gets closer to room temperature, it cools more slowly.
- Eventually it stops changing, once it matches the room.

TRANSLATING IT INTO SYMBOLS

In one sentence: the rate of cooling is proportional to how far the coffee's temperature is from the room's. Using T for the coffee's temperature, T_∞ for room temperature, and a positive constant k for how fast heat escapes:

$$\frac{dT}{dt} = -k(T - T_\infty)$$

The minus sign says: since $T > T_\infty$, the temperature is decreasing. This is Newton's Law of Cooling, built from common sense rather than memorized.

Population growth with limited resources

With unlimited food and space, a population grows in proportion to its own size — same structure as the plant: $\frac{dP}{dt} = rP$.

Real environments have a **carrying capacity** K — a maximum sustainable population. As P approaches K , growth should slow and eventually stop. A "braking" factor close to 1 when P is small, and close to 0 when P nears K , does the job:

PUTTING IT TOGETHER

$$\frac{dP}{dt} = rP \left(1 - \frac{P}{K}\right)$$

— the logistic growth equation from the slides, arrived at from two sentences of reasoning.

3 Two Words You'll See Constantly

TL;DR

Independent variable = what you choose. Dependent variable = what responds to it.

- The **independent variable** — the thing you're free to pick, usually time t or position x .
- The **dependent variable** — the thing being described, which changes because of the independent variable. In the plant example, h (height) depends on t (time).

Rule of thumb: whatever's being differentiated (as in $\frac{dh}{dt}$) is the dependent variable; whatever's on the bottom of that fraction is the independent one.

4 Why We Classify Equations At All

TL;DR

Type, order, and linearity each tell you which solving techniques will actually apply.

Before solving anything, it helps to know what kind of equation is being dealt with — the same way solving $2x = 6$ needs a different approach than solving $x^2 = 6$.

4.1 Type

- Only one independent variable (like t alone) → **ordinary** differential equation (ODE). This course focuses on these.
- Two or more (like both x and t) → **partial** differential equation (PDE) — used when something depends on both time and position, like heat spreading through a metal rod. That's a later course.

4.2 Order

Think about a car: position $x(t)$, velocity $= \frac{dx}{dt}$ (one derivative), acceleration $= \frac{d^2x}{dt^2}$ (two derivatives). Order looks at the *highest* derivative present — never how many appear in total, and never what power any of them is raised to.

EXAMPLE

$$\frac{d^2y}{dx^2} + 5 \left(\frac{dy}{dx} \right)^3 - 4y = e^x \rightarrow \text{2nd order}$$

$\left(\frac{dy}{dx} \right)^3$ is cubed, but $\frac{dy}{dx}$ itself is only a first derivative — the highest derivative present is $\frac{d^2y}{dx^2}$.

4.3 Linearity

A DE is **linear** if every term is just (some expression in x) times y or one of its derivatives, added together. Quick test: is y ever squared, stuck inside a **sin** or $e^{(\cdot)}$, or multiplied by its own derivative? If yes, it's **nonlinear**.

$$\frac{dy}{dx} = 3y \text{ — linear, just a constant times } y \quad | \quad \frac{dy}{dx} = y^2 \text{ — nonlinear, } y \text{ is squared}$$

WHY BOTHER?

Linear equations have well-developed, systematic solving methods — most of this course covers those. Nonlinear equations are often much harder; sometimes there's no exact formula, and the answer has to be approximated numerically.

5 What Does "Solving" (and "Checking") Actually Mean?

TL;DR

Checking a solution means substituting the function and its derivatives back in — and noting the interval where it's actually valid.

Just like checking whether $x = 3$ solves $2x + 1 = 7$ by plugging it in, a *function* is checked against a differential equation by substituting it — and its derivatives — back in.

THE SIMPLEST POSSIBLE CASE

$$\frac{dy}{dx} = 2x$$

What function has slope $2x$ everywhere? From basic integration, $y = x^2$ has derivative $2x$.

Check: $\frac{d}{dx}(x^2) = 2x \checkmark$.

A CASE FROM THE LECTURE

$y'' + y = 0$, try $y = \sin(x)$. Derivatives: $y' = \cos(x)$, $y'' = -\sin(x)$. Substitute back: $y'' + y = -\sin(x) + \sin(x) = 0 \checkmark$ — valid for every real x .

A CAUTIONARY CASE

$y'' = \frac{2}{x^3}$, try $y = \frac{1}{x}$. Derivatives: $y' = -\frac{1}{x^2}$, $y'' = \frac{2}{x^3}$ — matches ✓ — except at $x = 0$, where $y = \frac{1}{x}$ isn't defined. So this solution only holds on $(-\infty, 0)$ or $(0, \infty)$.

LESSON

A solution isn't just a formula that satisfies the equation — it also comes with an *interval* on which it actually makes sense.

6 Explicit vs. Implicit

TL;DR

Sometimes solving for y explicitly doesn't cleanly give one function, so the relationship is left tangled on purpose.

An **explicit** solution writes " $y =$ (a formula in x)". An **implicit** solution is a relationship between x and y that's deliberately left tangled.

DERIVING WHY A CIRCLE STAYS IMPLICIT

Take $x^2 + y^2 = 25$. Solving for y gives $y = \pm\sqrt{25 - x^2}$. That $\pm$ means this isn't *one* function — it's two (the top half of the circle, and the bottom half). Since a function can only give one output per input, it's usually left as $x^2 + y^2 = 25$ instead.

It can still be checked against a DE using implicit differentiation: $2x + 2y y' = 0 \Rightarrow y' = -\frac{x}{y}$. So $x^2 + y^2 = 25$ is an implicit solution of $\frac{dy}{dx} = -\frac{x}{y}$.

7 Families of Solutions

TL;DR

Every integration introduces an arbitrary constant, so solutions usually come in families.

STARTING FROM THE SIMPLEST DE THERE IS

$$\frac{dy}{dx} = 0$$

What functions have a derivative of zero everywhere? Any constant: $y = 5$, $y = -2$, $y = 100$ — in fact $y = C$ for any number C . This is a **one-parameter family of solutions**.

Bump it up slightly — $\frac{dy}{dx} = 2x$ — and the same idea gives $y = x^2 + C$: one constant, because a first-order equation is essentially one integration. A **second-order** equation needs **two** integrations, so it comes with **two** constants:

$$y'' - 2y' + y = 0 \quad \longrightarrow \quad y = c_1 e^x + c_2 x e^x$$

An n th-order differential equation typically has an n -parameter family of solutions — one constant for every layer of differentiation undone.

8 Systems: When Two Things Change Together

TL;DR

When two quantities affect each other's rate of change, both equations are needed together, not one at a time.

The classic picture: **rabbits and foxes** sharing an ecosystem. The rabbit population grows on its own but shrinks the more foxes are around; the fox population grows based on available rabbits but shrinks on its own over time. Neither equation stands alone:

$$\frac{dx}{dt} = f(t, x, y) \quad \frac{dy}{dt} = g(t, x, y)$$

A solution is a *pair* of functions satisfying both equations at once, on the same interval. A concrete (linear) example from the lecture:

EXAMPLE

System: $\frac{dx}{dt} = x + 3y, \quad \frac{dy}{dt} = 5x + 3y$

Solution: $x = e^{-2t} + 3e^{6t}, \quad y = -e^{-2t} + 5e^{6t}$

Checked the same way as always — differentiate each proposed function and confirm it matches the right-hand side of its own equation.



The Big Picture

A differential equation is a sentence about how something changes, translated into math. Solving it means finding the actual function that behaves that way — not a number, a whole curve. Before solving, the equation is classified (type / order / linearity) because that determines which tools apply. Once there's a candidate solution, it's checked by substituting it — and its derivatives — back in, noting the actual interval where it's valid. And because differentiating erases constants, solutions almost always come in families, with one arbitrary constant for every order of the equation.

Based on: ENGR 213, Lecture 1 (Sept 9, 2026), Prof. A. Haghighat M. — reference text: Zill, *Advanced Engineering Mathematics* (7th ed.), Section 1.1. Examples marked "let's derive" are original, built to illustrate the same ideas from the lecture.

Appendix: Lecture 1 Slide Figures

ENGR 213 · Introduction to Differential Equations · figures from the teacher's Lecture 1 slides, with a guide to reading each one

Where do differential equations appear?

- Cooling a hot object
How fast is the temperature changing? $\rho cV \frac{dT}{dt} = -hA(T - T_\infty)$
- Vehicle Motion
How fast is the velocity changing? $\frac{dx}{dt} = v$ $\frac{dv}{dt} = a$
- Chemical Reaction ($A \rightleftharpoons B$)
How fast is concentration of A changing? $\frac{dC_A}{dt} = -k_1 C_A + K_2 C_B$
- Population growth
How does a population change when resources are limited? $\frac{dP}{dt} = rP \left(1 - \frac{P}{K}\right)$

Different physical systems, different equations → But the common feature is: an unknown quantity is related to one or more of its derivatives.

Figure A1: Where do differential equations appear?, Lecture 1, slide 4.

Reading the slide: four unrelated systems, one pattern. Cooling ($\rho cV \frac{dT}{dt} = -hA(T - T_\infty)$), vehicle motion ($\frac{dx}{dt} = v$, $\frac{dv}{dt} = a$), a reversible reaction ($\frac{dC_A}{dt} = -k_1 C_A + K_2 C_B$) and limited population growth ($\frac{dP}{dt} = rP(1 - \frac{P}{K})$) all state **how fast something changes** in terms of its current value. The blue box is the definition you need: *an unknown quantity is related to one or more of its derivatives.*

Definitions and Terminology

- A **differential equation (DE)** is an equation containing the derivatives of one or more *dependent variables* with respect to one or more *independent variables*.

e.g.,: $\frac{dP}{dt} = rP \left(1 - \frac{P}{K}\right)$

- P is the dependent variable
- t is the independent variable
- $\frac{dP}{dt}$ is the derivative of y with respect to t

How do we solve such an equation for the unknown function $P = \phi(t)$?

- DEs can be classified by:

- Type
- Order
- Linearity

Figure A2: Definitions and terminology, Lecture 1, slide 6.

Reading the slide: in the logistic example P is the **dependent** variable (the unknown function), t the **independent** variable, and $\frac{dP}{dt}$ its derivative. Solving means finding the function $P = \phi(t)$. (The third bullet on the slide says "derivative of y "; read it as the derivative of P .) Every DE is classified three ways: **type** (ODE or PDE), **order** (highest derivative) and **linearity**.

□ Classification by Linearity

- An n th-order ODE, $F(x, y, y', \dots, y^{(n)}) = 0$, is linear in the variable y if F is linear in $y, y', \dots, y^{(n)}$.

$$a_n(x) \frac{d^n y}{dx^n} + a_{n-1}(x) \frac{d^{n-1} y}{dx^{n-1}} + \dots + a_1(x) \frac{dy}{dx} + a_0(x)y = g(x)$$

➤ Characteristics of a linear ODE:

- I. The dependent variable y and all its derivatives $y', y'', \dots, y^{(n)}$ are of the first degree (power of each term involving y is 1.)
- II. The coefficients $a_0, a_1, \dots, a_n$ of $y, y', \dots, y^{(n)}$ depend at most on the independent variable x .
- III. Nonlinear functions of the dependent variable or its derivatives such as $\sin(y)$ or e^y cannot appear in a linear equation.

Figure A3: Classification by linearity, Lecture 1, slide 10.

Reading the slide: a linear ODE has the form $a_n(x)y^{(n)} + \dots + a_1(x)y' + a_0(x)y = g(x)$. The three tests: (I) y and every derivative appear to the **first power**; (II) the coefficients depend **at most on x** ; (III) no nonlinear function of y such as $\sin y$ or e^y . A term like $x^3 y'''$ is fine; $y y'$ or $(y')^2$ is not.

□ Solution of an ODE

- Any function ϕ , defined on an interval I and possessing at least n derivatives that are continuous on I , which when substituted into an n th-order ordinary differential equation reduces the equation to an identity, is said to be a solution of the equation on the interval.

$$F(x, y, y', \dots, y^{(n)}) = 0$$



function $\phi(x)$ is a solution
of this ODE if

- I. $\phi(x)$ possesses at least n derivatives
 - II. $F(x, \phi(x), \phi'(x), \dots, \phi^{(n)}(x)) = 0$ (for all x in I)
- The interval I is also called the interval of definition or the domain of the solution and can be an open interval (a, b) , a closed interval $[a, b]$, an infinite interval (a, ∞) , and so on.

Figure A4: Solution of an ODE, Lecture 1, slide 13.

Reading the slide: a function ϕ is a solution on an interval I if (I) it has at least n derivatives that are continuous on I , and (II) substituting it turns the ODE into an identity **for every x in I** . The interval (the interval of definition) is part of the answer: $1/x$ solves $xy' + y = 0$ only on $(-\infty, 0)$ or $(0, \infty)$, never across 0.

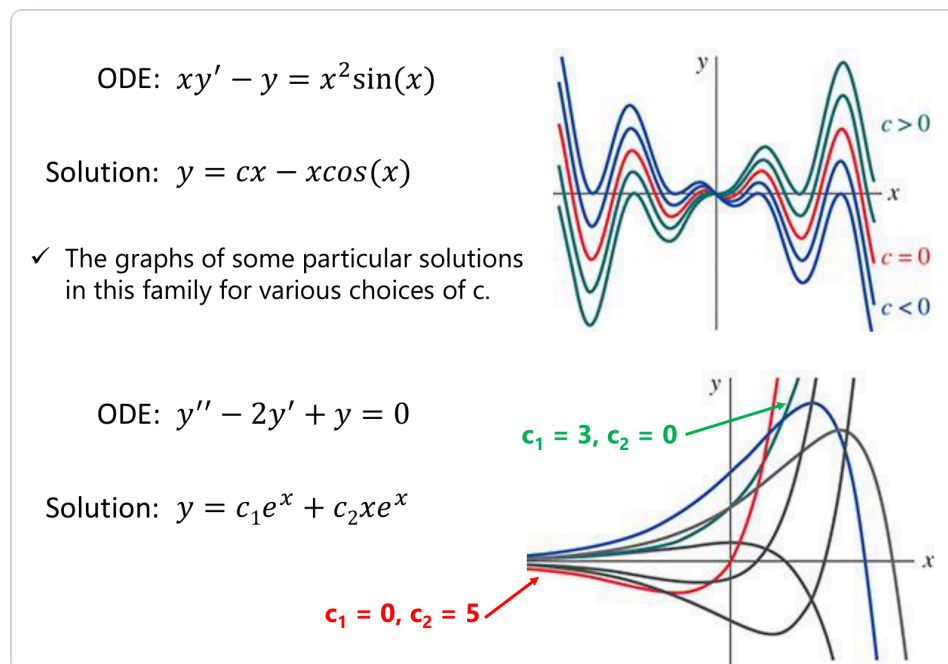


Figure A5: Families of solutions, Lecture 1, slide 19.

Reading the graphs: the first-order ODE $xy' - y = x^2 \sin x$ has the one-parameter family $y = cx - x \cos x$; green curves have $c > 0$, the red curve $c = 0$ and blue curves $c < 0$, and all of them pass through the origin. The second-order ODE $y'' - 2y' + y = 0$ needs **two** constants, $y = c_1 e^x + c_2 x e^x$, and each pair (c_1, c_2) picks one

curve: the green arrow marks $3e^x$ ($c_1 = 3, c_2 = 0$) and the red arrow marks $5xe^x$ ($c_1 = 0, c_2 = 5$). This is why an n th-order ODE needs n initial conditions.